Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 10

Quantifiers and Their Negations

Covers Chartrand & Zhang, Section 3.1

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Ask what the claim would cost

The four jobs

	To show TRUE	To show FALSE
$\forall x \in S, R(x)$	argue for *every* x	one counterexample
$\exists x \in S, R(x)$	one example	argue for *every* x

Decide which box you are in before you write anything. It tells you whether you owe an example or an argument — and half the errors in this section come from owing the wrong one.

Negation in one line

Flip the quantifier, negate the open sentence. With several quantifiers, flip them all, left to right, and negate the open sentence at the end.
And watch the domain. A statement true over $\mathbb{Z}$ can be false over $\mathbb{R}$. Read which one you are given.

Three mistakes to expect

1. **Overshooting.** The negation of "all are" is "at least one is not," never "none are."
2. **Leaving the quantifier alone.** $\sim(\forall x, R(x))$ is not $\forall x, \sim R(x)$.
3. **Swapping quantifier order.** $\forall x \exists y$ and $\exists y \forall x$ are different statements.

Part II: Practice Problems

Section A — True or false

1. Determine the truth value of each, with a one-line justification.

(a) $\forall n \in \mathbb{Z},\ n^2 \geq n$.

(b) $\exists n \in \mathbb{Z},\ n^2 = 2$.

(c) $\forall x \in \mathbb{R},\ x^3 = x$.

(d) $\exists x \in \mathbb{R},\ x^3 = x$.

(e) $\forall n \in \mathbb{Z},\ 3n + 1$ is even.

Section B — Negation

2. Write the negation of each, phrased the way a person would say it.

(a) Every student in this class has a laptop.

(b) Some prime number is even.

(c) $\forall n \in \mathbb{Z},\ 5n + 2$ is odd.

(d) $\exists x \in \mathbb{R},\ x^2 + x + 1 = 0$.

(e) There is no smallest positive real number.

Section C — Two quantifiers

3. For each, state whether it is true or false over $\mathbb{Z}$, and justify.

(a) $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z},\ ab = a$

(b) $\exists b \in \mathbb{Z}, \forall a \in \mathbb{Z},\ ab = a$

(c) $\exists a \in \mathbb{Z}, \forall b \in \mathbb{Z},\ ab = 0$

(d) $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z},\ a + b = 1$

4. Write the negation of each in symbols, then in words.

(a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ y^2 = x$

(b) $\exists a \in \mathbb{Z}, \forall b \in \mathbb{Z},\ ab \geq 0$

Section D — Writing

5. In five to eight complete sentences, explain why disproving a statement beginning $\forall$ requires only one counterexample, while disproving one beginning $\exists$ requires an argument covering the whole domain. Illustrate with two specific statements of your own.

Section E — Challenge (optional)

6. A statement of the form $\forall x \in S,\ P(x) \Rightarrow Q(x)$ can be false in only one way. Describe it precisely, say what a counterexample must therefore look like, and give a counterexample to: *for every integer n, if n^2 is even then n is a multiple of 4.*